Global Meteorological and Air Quality Analysis

ALY 6140: Python & Analytics Technology

Master Of Professional Studies In Analytics

Northeastern University, Toronto

Prepared By,

Group 4

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Introduction

As climate change accelerates and cities expand, understanding the connection between weather and air quality is more important than ever. Factors such as temperature, humidity, and wind directly influence pollution levels, which affect public health, agriculture, and the environment. Global meteorological and climate agencies, such as the World Resources Institute, rely on data analytics to guide environmental decision-making and policy formulation. These agencies need clear, actionable insights to anticipate poor air quality, prepare for extreme weather, and make smarter climate and sustainability decisions. In this study, we use global data to explore how weather and pollution interact, providing guidance for predicting temperature, rainfall, and air quality across different regions worldwide.

Dataset Overview

Source: "GlobalWeatherRepository" from Kaggle

Coverage: Multiple regions worldwide, several years, 72,225 cleaned records

Meteorological Variables: Temperature, humidity, wind speed, pressure, cloud cover, UV index

Air Quality Indicators: PM2.5, PM10, CO, SO_2, NO_2, O_{33}

Research Questions

1. What is the correlation between meteorological factors and air quality?
2. How effectively can temperature be predicted using meteorological and air quality variables, and which features contribute most to its variation?
3. Can meteorological and air quality variables accurately predict rainfall, and which factors are most influential?
4. Can air quality be classified into Good, Moderate, and Poor categories based on weather conditions and pollutant concentrations?

Methods Overview

Data Cleaning & Preprocessing:

- Outlier removal using IQR method
- Standardized numeric variables
- Encoded categorical features
- Removed underrepresented regions (<500 records)

Exploratory Analysis: Descriptive statistics, feature distributions, correlation analysis

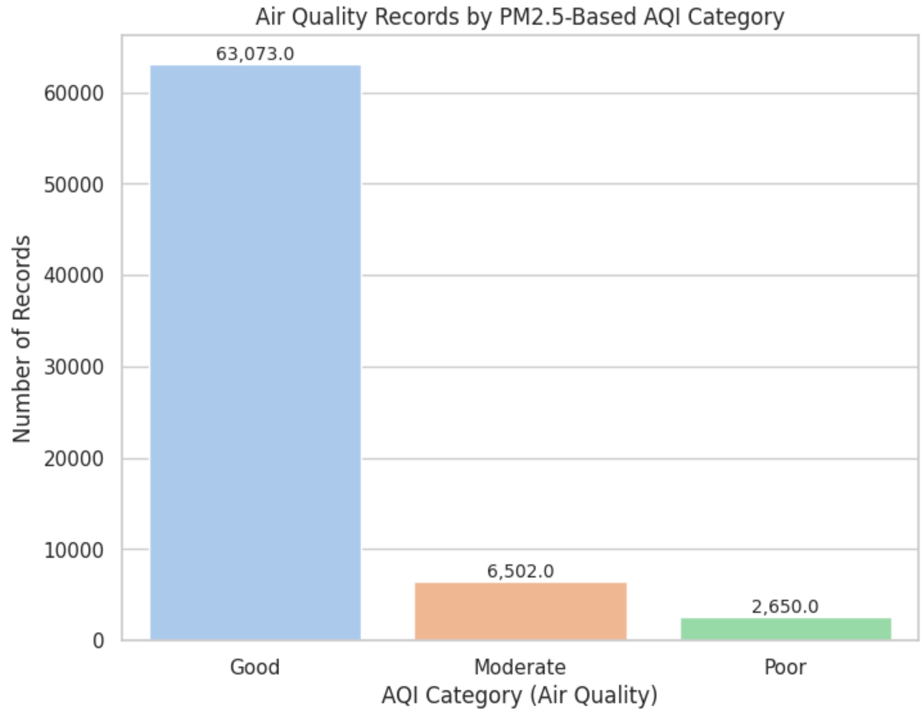
Predictive Models:

- Multiple Linear Regression → Temperature
- Logistic Regression → Rainfall
- XGBoost → Rainfall
- Random Forest → AQI classification

Descriptive Statistics

Summary Statistics for Selected Features:

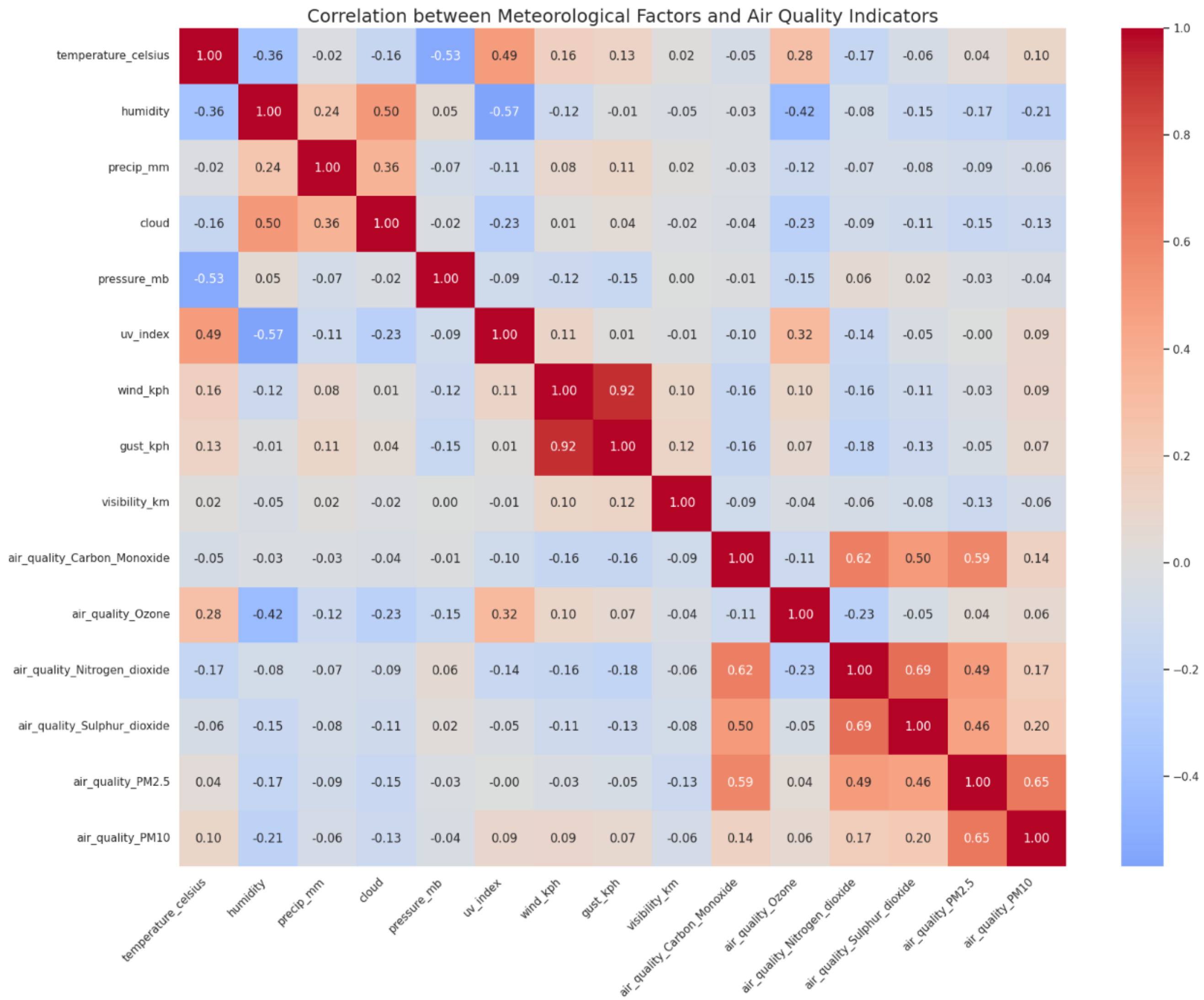
	count	mean	std	min	25%	50%	75%	max
temperature_celsius	98531.0	22.762491	8.889935	-24.900	18.10	24.900	28.300	49.20
humidity	98531.0	64.248003	24.170029	2.000	47.00	69.000	84.000	100.00
precip_mm	98531.0	0.141389	0.603095	0.000	0.00	0.000	0.030	42.24
cloud	98531.0	39.337366	33.746473	0.000	0.00	26.000	75.000	100.00
pressure_mb	98531.0	1013.984817	11.344283	947.000	1010.00	1013.000	1018.000	3006.00
uv_index	98531.0	3.865820	3.618322	0.000	0.50	3.000	6.900	16.30
wind_kph	98531.0	13.250973	12.636950	3.600	6.50	11.200	18.400	2963.20
air_quality_PM2.5	98531.0	26.055641	40.667722	0.168	7.40	15.010	29.785	1614.10
air_quality_PM10	98531.0	53.479783	163.458546	-1848.150	10.73	22.015	45.695	6037.29

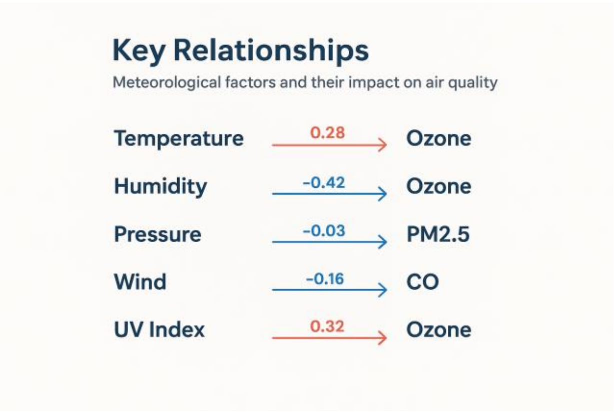


Proportion of Records per AQI Category:

	proportion
aqi_category	
Good	0.873285
Moderate	0.090024
Poor	0.036691

How do meteorological factors correlate with air quality indicators?





- High temperature and strong UV light → more ozone.
- High humidity and wind → less particulate matter and some gases.
- Weather directly affects air quality.

Predictive Analysis

How effectively can temperature be predicted using meteorological and air quality variables, and which features contribute most to its variation?

```
Multiple Linear Regression Results (Important Features Only):
```

	Feature	Coefficient
0	pressure_mb	-3.666
1	uv_index	2.564
2	wind_mph	5.256
3	cloud	-0.260
4	gust_mph	0.685
5	humidity	-1.206
6	wind_kph	-5.601
7	latitude	-1.282
8	air_quality_PM10	0.323
9	air_quality_PM2.5	-0.306
10	Intercept	23.659
11	RMSE	5.341
12	R^2	0.515

Multiple Linear Regression Model

These insights help improve temperature forecasts and guide better planning for heat and climate risks.

Can meteorological and air quality variables accurately predict rainfall, and which factors are most influential in determining rain events?

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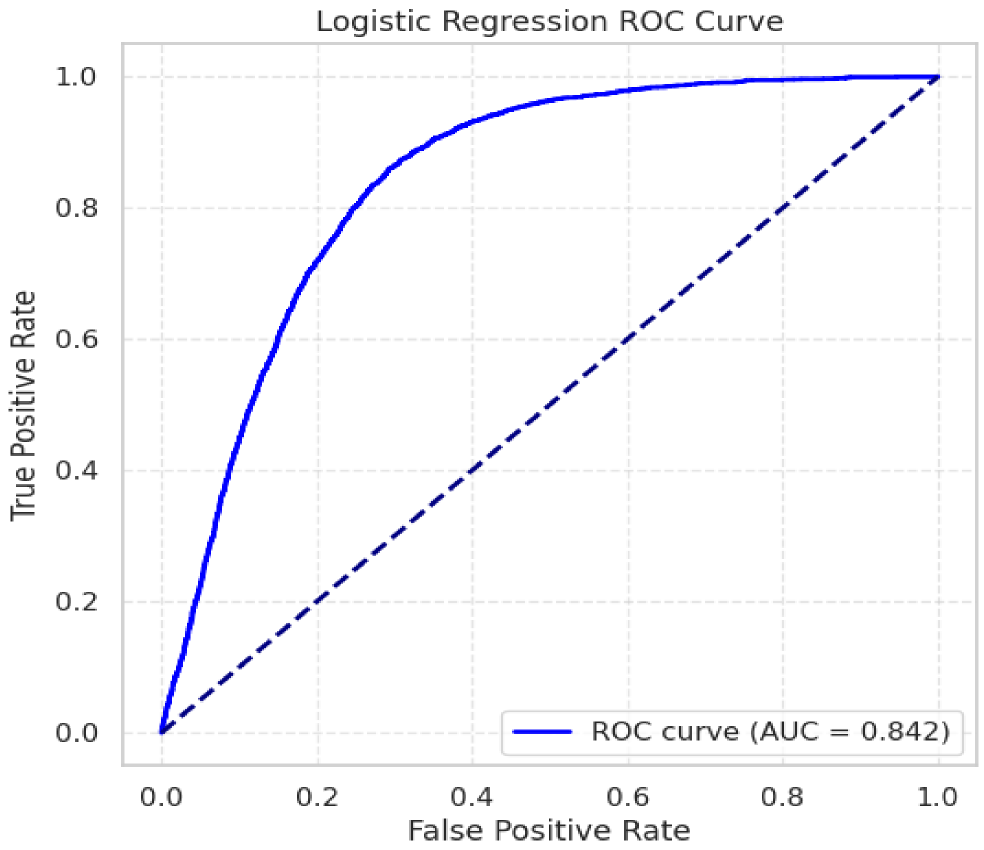
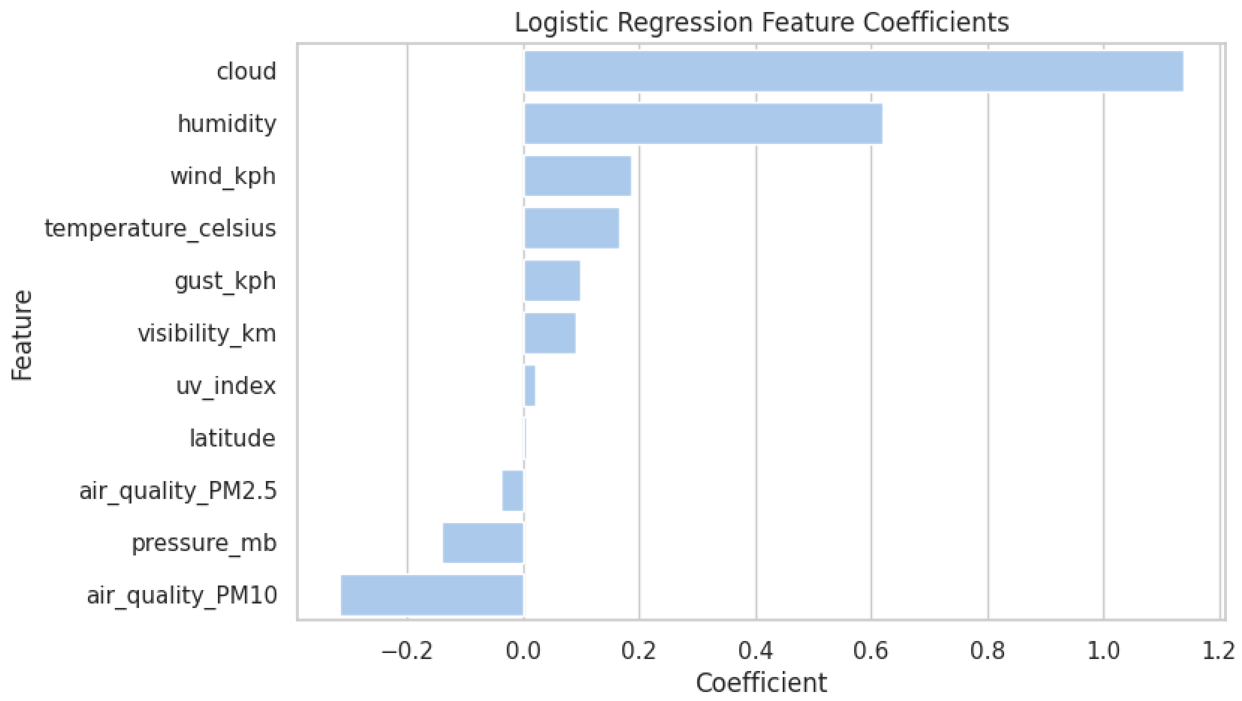
Logistic Regression Performance:
Accuracy: 0.756
Precision: 0.419
Recall: 0.813
F1 Score: 0.553
ROC AUC: 0.842
```





Can we enhance rain prediction performance using XGBoost, a gradient boosting algorithm, to capture nonlinear feature interactions.

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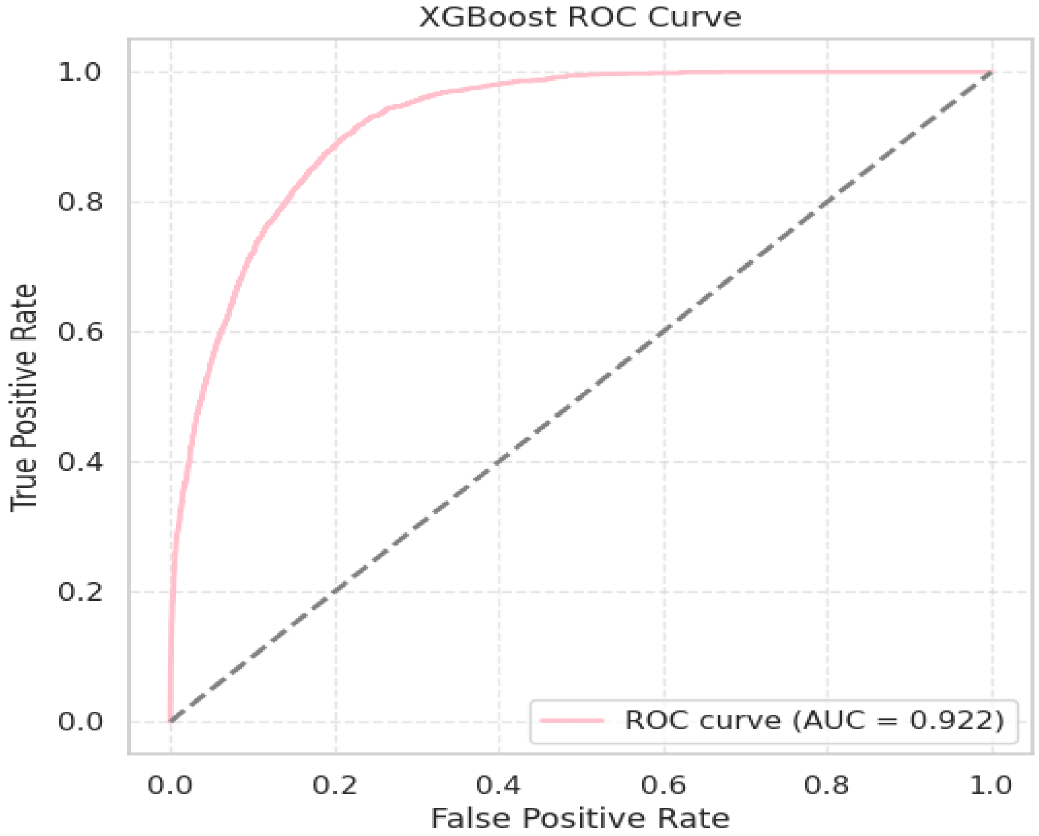
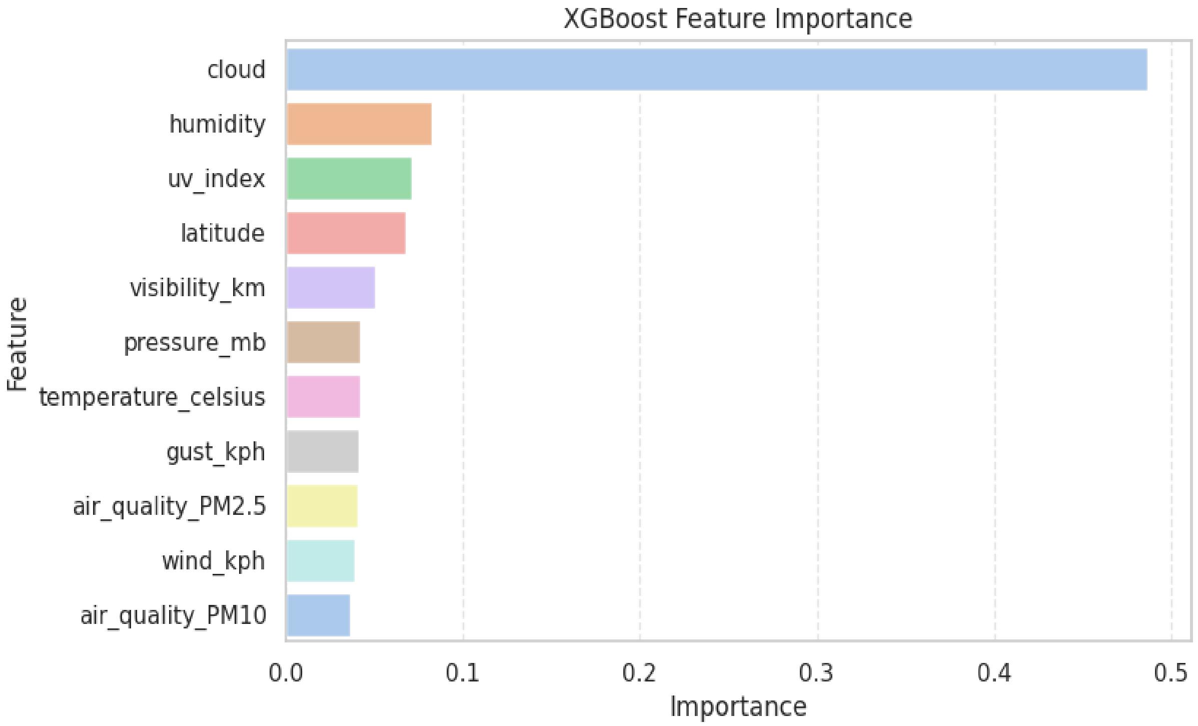
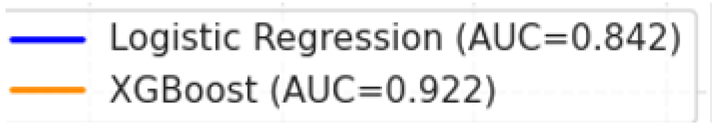
XGBoost Performance:
Accuracy: 0.876
Precision: 0.709
Recall: 0.568
F1 Score: 0.631
ROC AUC: 0.922
```







Can air quality be classified into Good, Moderate, and Poor categories based on weather conditions and pollutant concentrations?

Random Forest – Air Quality Classification

```
Class distribution after SMOTE:
aqi_category
Good        50458
Poor        50458
Moderate    50458
Name: count, dtype: int64
Random Forest AQI Classification Performance (with SMOTE):
              precision    recall  f1-score   support

        Good       0.98      0.90      0.94     12615
    Moderate       0.43      0.74      0.54      1300
        Poor       0.68      0.77      0.72       530

    accuracy                           0.88     14445
   macro avg       0.70      0.80      0.73     14445
weighted avg       0.92      0.88      0.90     14445
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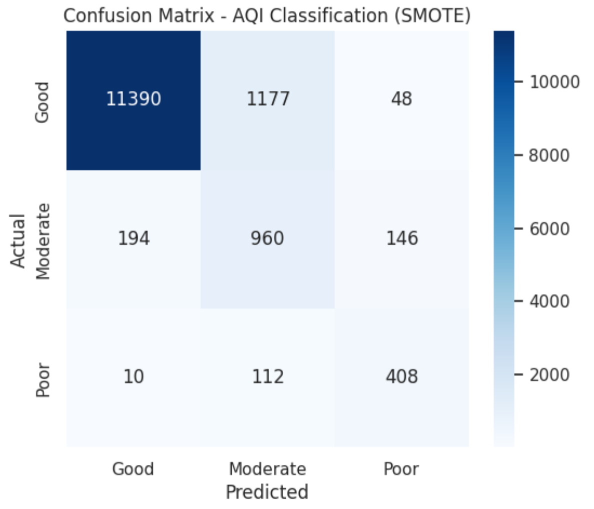

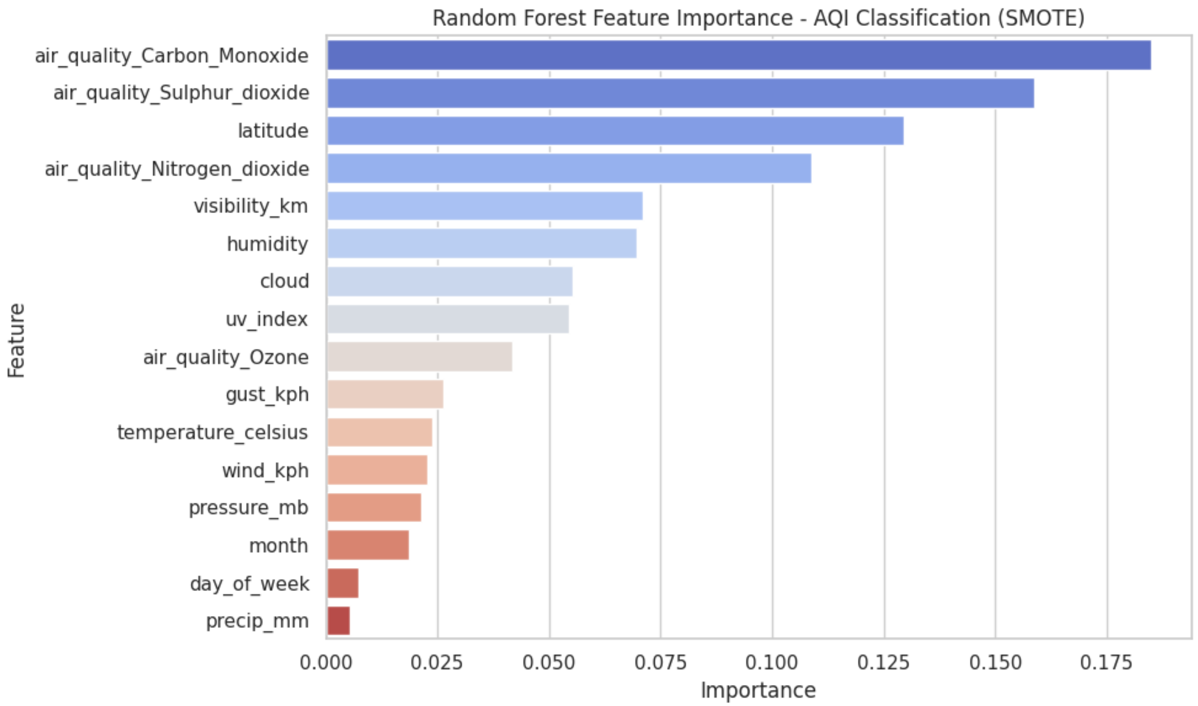

PM2.5 values categorized

Good: PM2.5 ≤ 50, Moderate: 51 ≤ PM2.5 ≤ 100, Poor: PM2.5 > 100 (WHO)

Which pollutants matter most, like CO, SO_2, and NO_2 , so global agencies can focus on reducing these emissions and issue timely health alerts.

Research Findings

Discussion – Insights for Policy and Action

Temperature Prediction: UV index, wind, pressure, and PM levels strongly influence daily temperature variations.

Rainfall Prediction: Logistic Regression captures most rain events; XGBoost handles complex patterns for higher confidence.

Air Quality Classification: CO, SO_2, NO_2, along with weather conditions, are primary drivers of poor air quality.

Implications: Understanding these patterns helps explain how weather and pollution interact globally and highlights areas for policy focus.

Conclusions & Recommendations

Temperature Prediction:

- Include UV index, wind, pressure, and PM levels in forecasts
- Support urban heat mitigation and disaster planning

Rain Forecasting:

- Use Logistic Regression for early warnings
- Use XGBoost for precise operational forecasts

Air Quality Management:

- Monitor CO, SO_2, NO_2 closely
- Issue alerts for Moderate/Poor AQI days
- Allocate resources regionally for pollution control

References

Data Source:

- Nelgiriyewithana, S. (2024). Global Weather Repository Dataset. Kaggle. https://www.kaggle.com/datasets/nelgiriyewithana/global-weather-repository/data

Climate & Policy Data:

- World Resources Institute. (2023). Global air quality and climate data. Retrieved from https://www.wri.org

Tools / Software:

- Scikit-learn developers. (2025). Scikit-learn: Machine learning in Python. https://scikit-learn.org

Thank You